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Class : S.E (Afternoon) A

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Assignment 01 :

• AI Chatbot :

• A. Purpose :

The purpose of an AI chatbot is to communicate with the users through natural language and provide automatic response to their queries.

It can also used for :

- Answering frequently asked questions.
- Providing Customer support.
- Providing assistance all the time.

• B. Uses :

The uses are follow as:

1. Customers — ask questions and receive support.
2. Students — ask education questions.
3. Employees — get help within company.

* Stakeholders:

They are the people or organization those are effected by as have an interest in the system.

- 1- Users/customers — Interact with Chatbot.
- 2- Business/organization — Use Chatbot to improve.
- 3- Developers — deploy and maintain Chatbot.
- 4- AI/Cloud service — Provide AI model and infrastructure.

* D. Functional Requirements:

It means what a chatbot do.

1. User Input:

The system shall allow user to enter question or message.

2. Understand queries:

The chatbot shall allow user to enter and understand natural language.

3. Generate Responses:

The chatbot generate relevant response.

4. Conversation History:-

The System Should maintain conversation content during a session.

5. Error Handling:-

The chatbot Should provide an inappropriate.

1. Performance:-

The chatbot Should response user within a reasonable amount.

2. Availability:-

The System Should Available 24/7 with minimal downtime.

3. Security:-

The system Should be protected from unauthorized data.

4. Conversation History:-

The System should be able to convert the data and show history.

5. Useability:-

The system should have easy to use interface.

6. Risks:-

1- Incorrect Responses

The AI may provide false and misleading information.

2- Privacy & Security:-

Sensitive user information could be exposed.

3- Misuse:-

User may rely on Chatbot for decision that require human expertise.

* G1. Quality Attribute:-

1. Reliability:-

The Chatbot should consistently provide appropriate responses.

2. Accuracy:-

The system should understand user queries and provide answers.

3. Usability:-

User should be able to interact with chatbot easily.

H. Where AI could be useful:-

AI is the core technology for an AI chatbot. It can be used for:-

- Natural language processing (NLP):-

Understand

human language.

- Intent Recognition:-

Determine what

user wants.

- Response generation:-

Generating

natural language answers.

I. What could be wrong if AI is incorrectly used:-

If AI is implemented or used incorrectly then:-

- The chatbot may provide false information.
- It may provide inappropriate response.
- It may misunderstand user's question.
- Private or sensitive information could be exposed.